

How Length of Stay and Billing Vary Across Hospital Admissions

A SQL and Power BI analysis of a synthetic healthcare dataset

Final Career Simulation | Data Analytics and Visualization

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Eric led data preparation, cleaning, validation, and SQL analysis. Wil built the Power BI dashboard.

Dataset: Eduardo Licea's modified healthcare dataset · July 2026

Project Summary

We analyzed a healthcare admissions dataset to find how length of stay and billing vary across admissions, and whether that variation is associated more with a patient's medical condition or with how the patient was admitted. The work was split cleanly. I handled the data side end to end, profiling the raw file, cleaning and standardizing it, validating the result against the untouched raw file, and preparing an analysis-ready dataset. My partner, Wil Schmiedeknecht, built the interactive Power BI dashboard on top of that dataset.

The source is Eduardo Licea's modified healthcare dataset, a synthetic file of 55,500 admission records across 16 columns. One point shapes how every result should be read. Each row is an admission, not a unique patient, because the file carries no permanent patient identifier, so every count in this report is a count of admissions.

The clearest pattern is in length of stay by condition, which ranges from 2.50 days for Flu admissions to 54.42 days for Alzheimer's admissions. Cancer shows the highest average billing of any condition. When condition is held constant, average billing differs little across admission types, with the Emergency-to-Elective spread inside a condition running only negative 2.51% to positive 3.46%. Condition is where this analysis found the variation worth building around, which makes it the sensible starting point for the dashboard.

The analysis rests on a validated pipeline. I kept the raw data untouched as a baseline, built the cleaned copy separately, and reconciled the two after every change. The cleaned table preserved all 55,500 rows and produced a net billing difference of \$0.00 against the raw table, and it passed additional checks on encounter-ID uniqueness, value ranges, length of stay, dates, and quality exceptions. Matching the row count and the billing total does not by itself prove every individual value is unchanged, but together these checks give me strong confidence the cleaning did not drop or distort the data.

This report is one of three deliverables for the project. It is submitted alongside a visual evidence file of dashboard screenshots and a project ZIP containing the SQL scripts and the Power BI file.

Methodology

I ran the project in the order the work actually happens: frame the questions, profile the raw data, clean and standardize it, validate the result, prepare a lean handoff table, run the analysis

queries, and hand the clean file to the dashboard. All of the SQL work was done in SQLite through DB Browser. Because a bootcamp project has no live database server, the SQL stage and the Power BI stage are bridged by a single manual export. I cleaned in SQL, exported the analysis-ready table as a CSV, and that CSV was the only thing the dashboard read.

Choosing the dataset. Before committing to a source, I compared the original Prasad Patil healthcare dataset with Eduardo Licea's modified version. Both files contain 55,500 rows. Comparing them position by position, Doctor and Room Number matched on all 55,500 rows, Name matched on 33, and Hospital and Billing Amount matched on none. That confirms the two files are not the same file, but it says nothing about which one is more accurate, since both are synthetic. I selected Licea's version for a practical reason. It adds a Length of Stay field that the project analysis requires, which the original does not carry. The choice was about having the column the questions depend on, not about which file produced more interesting results.

Three tables, one dataset. I structured the database as three tables rather than editing one in place. The healthcare_raw table holds the original imported data, and I never changed a value in it, so it stays as a baseline to reconcile against. The healthcare_clean table is the full cleaned copy where every fix lives. The healthcare_analysis_ready table is a trimmed copy that carries only the fields the dashboard needs. It is one dataset at three stages: original, cleaned, and ready for handoff.

Profiling before touching anything. I confirmed the raw table had 16 columns and 55,500 rows and previewed records to check the columns lined up. I checked how each column imported and applied explicit casts later in the cleaning so the types are enforced the same way regardless of what the import inferred. I checked all 16 columns for nulls, empty strings, and space-only cells and found none, and I checked the text fields separately for stray leading or trailing spaces and found none. I profiled the categorical fields for misspellings, inconsistent capitalization, and duplicate labels, and they were largely consistent. Checking the numeric ranges, age ran 5 to 90, billing from about \$500 to just under \$100,000, room numbers 101 to 500, and length of stay 1 to 89 days, and a separate query for impossible values such as negative billing, negative stay, invalid rooms, or ages out of range returned nothing. On the dates, admissions ran from May 8, 2019 to May 7, 2024, and discharges extended to July 28, 2024, which fits because some stays continued past the last admission date. Every date parsed, and no discharge fell before its admission. I flagged May 2019 and May 2024 as partial months so their smaller totals are not misread as a real decline.

The file came with a Length of Stay column, but I did not trust it on sight. I recalculated it from the two dates with SQLite's julianday function, and all 55,500 supplied values matched what I computed, with zero mismatches. I checked for exact duplicates across all 16 columns rather than on name alone, since two admissions sharing a name are not necessarily the same record, and found no duplicate groups. Rather than deleting the largest values as outliers, I reviewed them. The highest billing sat in Cancer admissions and the longest stays in Alzheimer's admissions, both internally consistent, so I kept them. An extreme value is not automatically an error.

The one cleaning decision that mattered. Most of the file was already clean. The one substantive fix was in the medication column, which listed both Tamiflu and Oseltamivir. Those

are the same drug, Tamiflu being the brand name for oseltamivir, so treating them as two separate labels splits one active ingredient into two. There were 2,326 Tamiflu records and 2,329 already labeled Oseltamivir. Rather than overwrite the original labels, I carried both versions into the handoff. A medication_source column preserves the original split, Tamiflu 2,326 and Oseltamivir 2,329, and a medication column normalizes Tamiflu into Oseltamivir, giving Tamiflu 0 and Oseltamivir 4,655. That lets Wil compare the original brand and generic labels or group by active ingredient, so the original distinction stays available rather than being lost.

Building the clean table. I built healthcare_clean with two stacked common table expressions. The first standardizes the column names, trims whitespace, handles nulls, casts the types, formats the dates, and applies the medication standardization. The second calculates length of stay from the cleaned dates. I created an encounter_id from SQLite's rowid, since the source has no unique key, and I treat it as a snapshot row identifier for this database, not a real patient or healthcare ID. Along the way I added a few analytical fields: billing_per_day, an abnormal-test flag, a length-of-stay audit flag, and admission year and month. I am deliberate about calling billing_per_day a billing measure rather than a cost or efficiency measure, because billing is the amount charged, which is not the same as what the care cost or what was paid.

Checking the cleaning. This is the part of the work I lean on hardest. After building the clean table, I reconciled it against the raw table. Both had 55,500 rows, so every record was preserved. Total billing came to \$1,211,844,977.27 on both, a net difference of \$0.00. On top of that, all 55,500 encounter IDs were unique, the value ranges matched, there were no length-of-stay mismatches, the dates all held, and the final quality-exception query returned nothing. A matching row count catches lost rows and a matching billing total catches large shifts in value, but the two together do not prove that every individual cell is unchanged. What they do, alongside the uniqueness, range, length-of-stay, date, and quality checks, is give strong evidence that the cleaning preserved the data rather than damaging it.

Preparing the handoff. I then built healthcare_analysis_ready, the lean table for Power BI. It carries the analytical fields and leaves out columns the planned dashboard did not need, including name, doctor, room number, and the intermediate length-of-stay fields. Those columns were not required for the dashboard, so they were omitted from the lean handoff, and they stay available in healthcare_clean and healthcare_raw. I reconciled the lean table against the full clean table, both at 55,500 records, so no admission was dropped in the trim. This is the data-modeling and preparation layer of the project, and it keeps a clean separation between the cleaning work and the dashboard. Conditions and admission types are the dimensions a reader slices by; billing and length of stay are the measures.

Running the analysis. With the analysis-ready table set, I answered the project questions in SQL as a set of aggregation queries. Two of them independently recompute the headline numbers, the average stay and the billing totals, as a check on my own earlier results. Several use a CROSS JOIN to attach the dataset-wide averages onto every condition, so each condition is compared against the whole dataset inside the same query. The results are in the next section.

The handoff to Power BI. The analysis-ready table was the single input to the visualization layer. I exported it as a CSV and handed it to Wil, who built the dashboard in Power BI. Building

it from the pre-cleaned, pre-modeled table means the visualization does not repeat the SQL cleaning and cannot drift from it. The dashboard's visualizations, KPI views, and interactive filtering are captured in the accompanying visual evidence file and the Power BI file.

Analytical integrity and tooling. I used AI tools as a drafting and review assistant, to help organize SQL, explain syntax, and check terminology, but never as a source of results. Every query was run in DB Browser against the actual database, and I read, interpreted, and recorded each result from my own screen. I kept a separate markdown file as an audit log of what I checked, what it returned, why it mattered, and whether it passed. That discipline caught real errors, including a case where a tool mislabeled billing as cost and a case where a result was overstated, and it is why I can defend any number in this report against the database it came from.

Key Findings and Insights

At the portfolio level, the dataset holds 55,500 admissions with an average stay of 17.84 days, total billing of about \$1.212 billion, and average billing of \$21,835.04 per admission.

Metric	Value
Admissions	55,500
Average length of stay	17.84 days
Total billing	\$1,211,844,977.27
Average billing per admission	\$21,835.04

By admission type, Emergency is the largest group but not a majority, at 36.22% of admissions, followed by Elective, Urgent, and Routine.

Admission type	Share of admissions
Emergency	36.22%
Elective	23.93%
Urgent	21.29%
Routine	18.56%

Length of stay and billing vary by medical condition. This is the central pattern, and the range in length of stay is large. Average length of stay runs from 2.50 days for Flu admissions to 54.42 days for Alzheimer's admissions, with Cancer at 36.54 and Heart Disease at 26.86. On billing, Cancer has the highest observed average of any condition, and the highest average for any condition-and-admission-type combination is the Cancer and Emergency group at \$65,225.87 per admission.

Medical condition	Average length of stay
Alzheimer's	54.42 days

Medical condition	Average length of stay
Cancer	36.54 days
Heart Disease	26.86 days
Flu	2.50 days

The conditions above span the range, from the longest average stay (Alzheimer's) to the shortest (Flu).

To compare admission types on a like-for-like basis, I held condition constant and compared average billing for Emergency against Elective inside each condition. The difference ran from negative 2.51% to positive 3.46%, with Emergency higher in three conditions and lower in five and no consistent direction. Within a condition, average billing looks about the same whether the admission was Emergency or Elective. This comparison was on average billing only. I did not run the equivalent comparison for length of stay, so I make no claim about how admission type relates to length of stay, and I did not test any of these differences for statistical significance, so they are descriptive.

How stay and billing per day line up by condition. Looking at length of stay against billing per inpatient day, Cancer and Heart Disease sit above the overall averages on both. Alzheimer's shows the opposite mix, longer stays but lower billing per day, which makes it more of a length-of-stay pattern than a billing one. Diabetes, Obesity, and Asthma run shorter stays with higher billing per day. These are descriptive profiles, not measures of efficiency or cost.

Volume and intensity are mostly separate. The five highest-volume conditions make up 62.87% of all admissions, so most admissions fall into a handful of conditions. Among those five, Cancer is the only one that sits above the overall averages for both stay and billing. Flu, Diabetes, Obesity, and Asthma all fall below both. The conditions with the highest admission counts are mostly the lighter ones on stay and billing, and Cancer is the exception that is both common and high on both measures.

The supporting factors turned out to be low-signal. The project scoped test results and medication as secondary factors, and the honest result is that neither separates the data much. Abnormal-test rates fall in a narrow band from 49.15% to 50.46% across conditions, close to a coin flip everywhere, so that field gives almost no separation between conditions. Test Results here is a dataset label, not a clinical outcome, and I do not read it as one. For medication, grouping by active ingredient gives 23 condition-medication groups, and the differences in stay and billing within each group are small, which is limited descriptive separation. That is not evidence that medication has no effect, because this dataset cannot establish effectiveness or causation. Ruling these two factors out as weak separators is itself a finding, and it is worth stating rather than forcing a relationship that is not there.

Final Recommendations

The findings point to a clear way to use this data and a few honest limits on it.

Segment by condition first. Every operational and financial question in this dataset should start from medical condition, with admission type as a secondary descriptor. Condition is where the analysis found the variation, in the wide length-of-stay range and in Cancer's high average

billing, while within a condition the Emergency-to-Elective billing difference is small. Condition should be the primary filter in any view. Admission type is worth showing to describe the arrival mix, but on the billing comparison I ran it adds little once condition is known.

Focus attention on Cancer and Alzheimer's. These two conditions stand out, for different reasons. Alzheimer's has the longest average stay of any condition, so it is the one to watch for long individual admissions and discharge planning, though it is not among the highest-volume conditions. Cancer has the highest average billing and one of the longer average stays, and it is also one of the five highest-volume conditions, which puts it at the intersection of high volume and a high position on both stay and billing. That combination makes Cancer the segment that most rewards a close look in capacity planning and financial review.

Keep the supporting factors as context, not headline factors. Because abnormal-test rates and medication show so little separation between conditions, the dashboard should not build its main story on either. They belong as supporting context a reader can drill into, not as headline factors, and presenting them that way keeps the dashboard honest about where the separation actually is.

Limitations. A few limits should travel with every number here. Each row is an admission, not a unique patient, since the file has no permanent patient identifier, so all counts are counts of encounters. Billing is the amount charged, which is not hospital cost, payment, or revenue. Test Results is a dataset label, not a clinical outcome. The data is synthetic, so none of these figures is a real clinical or financial benchmark. Every result reported here is a description of the data, not a causal or significance-tested finding. The data covers about five years, from May 2019 through May 2024, and the first and last months are partial, so May 2019 and May 2024 should be excluded or clearly labeled in any monthly trend comparison.

What would strengthen the analysis next. Four additions would take this closer to a real operational analysis. A permanent patient identifier would let the analysis follow patients rather than only admissions, and make readmissions visible. True cost and payment fields would separate billing from cost and open up margin. Fuller time coverage, with complete first and last months, would support proper monthly trend analysis. A severity or acuity measure at admission would help account for the variation in stay and billing within a single condition, rather than only describing it. Naming where the analysis stops reads as command of the material, and it tells a reader how far these numbers can be pushed.